

MATH 4175 - Cryptography Project 1
Project 1 - Division Algorithm

Group 5

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Finding gcd of 768336 and 78192

	u1	v1	u2	v2	u3	v3
q						
	1	0	0	1	768336	78192
0	0	1	1	-9	78192	64608
9	1	-1	-9	10	64608	13584
1	-1	5	10	-49	13584	10272
4	5	-6	-49	59	10272	3312
1	-6	23	59	-226	3312	336
3	23	-213	-226	2093	336	288
9	-213	236	2093	-2319	288	48
1	236	-1629	-2319	16007	48	0
6						

gcd(768336, 78192) = 48
x = 236
y = -2319
Verification:
 $768336(236) + 78192(-2319) = 48$

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Finding gcd of 494752 and 296864

	u1	v1	u2	v2	u3	v3
q						
	1	0	0	1	494752	296864
0	0	1	1	-1	296864	197888
1	1	-1	-1	2	197888	98976
1	-1	2	2	-3	98976	98912
1	2	-3	-3	5	98912	64
1	-3	4637	5	-7728	64	32
1545	4637	-9277	-7728	15461	32	0
2						

gcd(494752, 296864) = 32

$x = 4637$
 $y = -7728$
 Verification:
 $494752(4637) + 296864(-7728) = 32$

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Finding gcd of 17601969 and 2364768

	u1	v1	u2	v2	u3	v3
q						
0	1	0	0	1	17601969	2364768
7	0	1	1	-7	2364768	1048593
2	1	-2	-7	15	1048593	267582
3	-2	7	15	-52	267582	245847
1	7	-9	-52	67	245847	21735
11	-9	106	67	-789	21735	6762
3	106	-327	-789	2434	6762	1449
4	-327	1414	2434	-10525	1449	966
1	1414	-1741	-10525	12959	966	483
2	-1741	4896	12959	-36443	483	0

$\text{gcd}(17601969, 2364768) = 483$
 $x = -1741$
 $y = 12959$
 Verification:
 $17601969(-1741) + 2364768(12959) = 483$